MILITARY SPECIFICATION SHEET

# LIGHT, LANDING, AIRCRAFT, RETRACTABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Complete requirements for procuring the light described herein shall consist of this document and the issue in effect of MIL-L-81174.

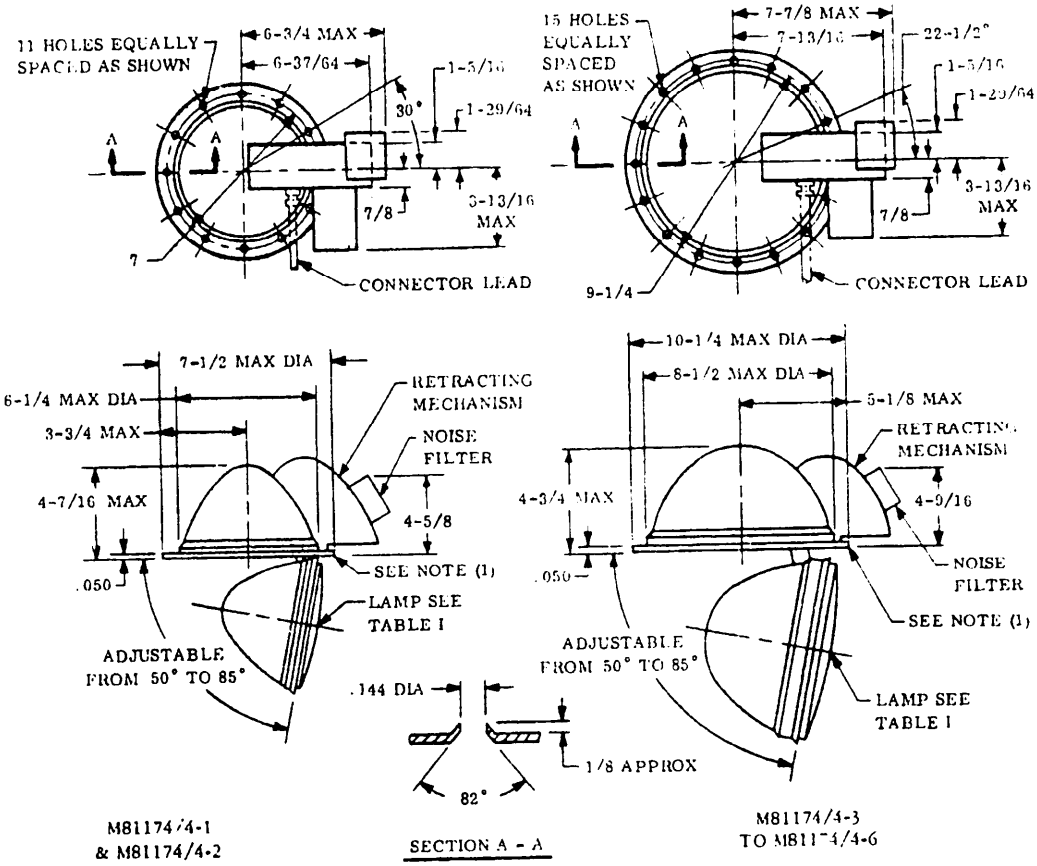



FIGURE 1. AIRCRAFT RETRACTABLE LANDING LIGHT



FSC 6220

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NOTES:

1. At this point the fixture shall accommodate a total thickness of skin and doubler plate of 0.100 inch.

2. Dimensions in inches unless otherwise specified. Tolerances: Decimals ±0.010, fractions ±1/32, angles ±1/2 degree. Dimensioning and tolerancing shall be in accordance with USASI Y14.5-1966.

3. Mounting rings to be used at contractor's option.

4. Part Number M81174/4-5 is inactive for design. Applications formerly requiring an M81174/4-5 Landing Light either have been or should be converted to the use of an M81174/4-4 Landing Light.

TABLE I

| PART NUMBER | MOUNTING HOLE CIRCLE DIAMETER (INCHES) | LAMP | SUPERSEDES (REF.) |
|---|---|---|---|
| M81174/4-1 | 7 | MS25241-4553 | AN3095-7 |
| M81174/4-2 | | WITHOUT LAMP | AN3095-4A |
| M81174/4-3 | 9-1/4 | WITHOUT LAMP | AN3095-6A |
| M81174/4-4 | | MS25242-4559 | AN3095-8, -9, -11 M81174/4-5 |
| M81174/4-5 | | MS25242-4569 | AN3095-10 |
| M81174/4-6 | | MS25242-Q4559 | - |

REQUIREMENTS:

Dimensions and configuration: See Figure 1.

Extending limit: Fully extended operating position shall be adjustable from not more than 50 degrees to not less than 85 degrees. The fully extended operating position shall be set at 73 ±3 degrees for shipment.

Braking: Within plus or minus 2 degrees under air load of 15 pounds.



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inventory, and the number of items to be tested as specified in # 1. The

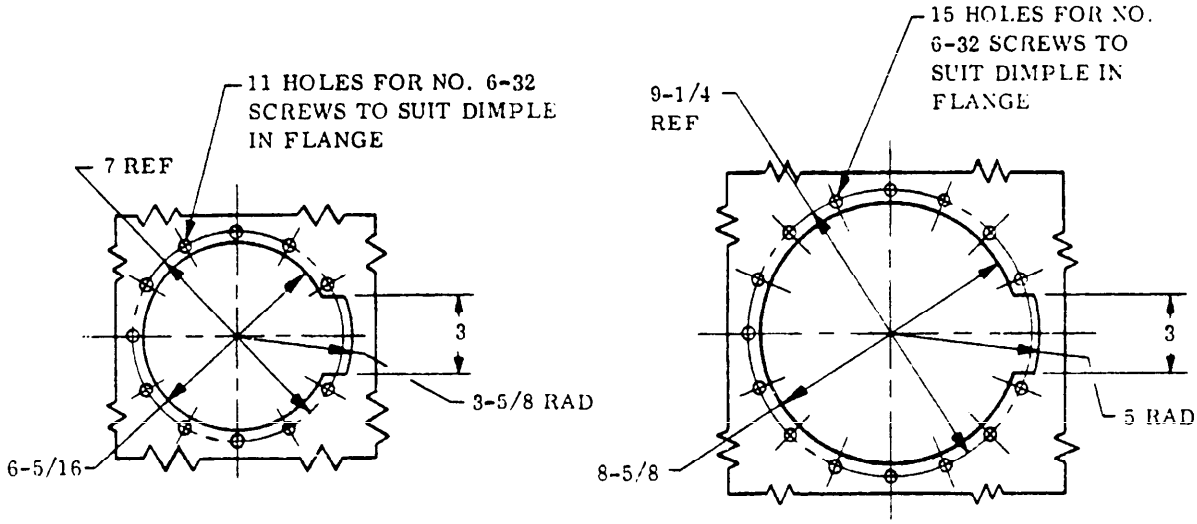



FIGURE 2. DESIGN FOR INSTALLATION AREA



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inventory, and the number of items to be tested as specified in 4.3. The

REQUIREMENTS (Continued):

Lamp switch: A switch shall be provided in the lamp circuit, which will close the circuit in all positions of the light or which can be so adjusted that the lamp circuit is automatically opened when the light is in any position between fully retracted and 10 degrees extended, and closed in all other positions.

Lamp: Light shall accommodate a PAR 64 or PAR 46 lamp.

Terminals: The wiring from the motor, lamp, and limit switch shall terminate in an electrical connector conforming to MS3102A-18-5P.

Connector connections: The electrical connector shall be connected as follows:

Terminal "A" - Motor retract
Terminal " B"- Motor extend
Terminal " C"- Lamp, positive terminal

Weight: 6.5 pounds maximum, including the lamp.

Qualification tests: Qualification tests shall be in accordance with MIL-L-81174 except as specified herein.

Low temperature test: The low temperature test shall be in accordance with MIL-L-81174. After stabilization at -54° C the light shall be operated at an ambient temperature of -54° C and the time required for the motor to extend shall be determined. During operation the closed circuit voltage shall be 28.0 VDC and a simulated air load of 15 pounds shall be applied as specified for the Life Test. The time to extend shall not exceed 17 seconds.

High temperature test: The high temperature test shall be in accordance with MIL-L-81174. While at an ambient temperature of 71° C the light shall be operated and the time required for the motor to extend shall be determined. During operation the closed circuit voltage shall be 28.0 VDC and a simulated air load of 15 pounds shall be applied as specified for the Life Test. The time to extend shall not exceed 17 seconds.

Humidity test: Upon conclusion of the test, the time required for the motor to extend the light shall be determined with a 15 pound load, with a closed circuit voltage of 28 VDC at the input terminals, and with an ambient temperature of approximately 20° C. The time to extend shall not exceed 17 seconds.

Sand and dust: Upon conclusion of the test, the time required for the motor to extend the light shall be determined with a 15 pound load, with a closed circuit voltage of 28 VDC at the input terminals and with an ambient temperature of approximately 20° C. The time to extend shall not exceed 17 seconds.



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REQUIREMENTS (Continued):

Life test: The life test shall be as specified in MIL-L-81174 except that operation shall be for 5,000 cycles at the rate of 25 to 30 cph. At least one of the first 10 cycles shall be operated at 30 VDC. At the completion of the test the extension time shall not exceed 17 seconds.

Acceptance tests: Acceptance test shall be as specified in MIL-L-81174, but shall also include a sampling test.

Sampling test: The sampling test shall consist of individual tests, the high temperature test and the low temperature test of MIL-L-81174. At the conclusion of the temperature tests the extension time shall not exceed 17 seconds with a 15 pound load, a closed circuit voltage of 28 VDC, and with an ambient temperature of approximately 20° C. The sample size shall consist of one lamp selected at random from each lot consisting of 1000 lights or fraction thereof.

Supersession data: This specification sheet in conjunction with MIL-L-81174 supersedes MIL-L-8346 and AN3095.

Custodians:
Navy - AS
Air Force - 82
Army - AV

Review Activities:
Navy -
Air Force -
Army -

Preparing Activity.
Navy - AS
(Project No. 6220-0209)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE MIL-L-81174/4A LIGHT, LANDING, AIRCRAFT, RETRACTABLE

| NAME OF ORGANIZATION AND ADDRESS | CONTRACT NUMBER |
|---|---|
| | MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT |

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

| SUBMITTED BY (Printed or typed name and address - Optional) | TELEPHONE NO. |
|---|---|
| | DATE |

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